

A Typed Arithmetic L^2 Interface for Four-Packet Gate-B Reassembly

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Abstract

The preceding four-packet analysis separates a signed reassembly gain from the arithmetic estimate that would have to produce it. This note makes that separation typed. Let A_X be an operator from a source Hilbert space into $\ell^2(I_X)$ and suppose $\|A_X\|_{2 \rightarrow 2} \leq KX^{-\sigma}$. If four packets have diagonal energy D , reassembled energy G , and normalized gain input $G/D \leq Q_X$, then the exact interface is

$$\|A_X S\|_2^2 \leq K^2 X^{-2\sigma} Q_X D.$$

Combining the two-term budget from TPC-280, $Q_X \leq BX^{-\gamma} + (\ell/d)X^{-\delta}$, with an upper source envelope $D \leq d_+ X^a$ gives the collapsed exponent $a - 2\sigma - \min(\gamma, \delta)$. Scalar readouts contract by Cauchy–Schwarz. We then prove an exact attachment obstruction: for the same nonzero packet sum in $\mathbb{R}^2$, two rank-one functionals have equal operator norm but give respectively full and zero attachment. Thus packet geometry and an L^2 norm do not identify the arithmetic attachment. Four exact rational packet witnesses, four interface cases, and the twelve-row TPC-280 transfer are independently certified. The operator estimate and nondegenerate attachment for the literal growing prime source remain open; no fixed-power credit or twin-prime conclusion is claimed.

1 Typed source interface

Let $X \geq 1$ be a scale, $\mathcal{H}_X$ a Hilbert space, and I_X an index set. Write

$$A_X : \mathcal{H}_X \longrightarrow \ell^2(I_X)$$

for a linear operator. Four source packets $V_0, \dots, V_3 \in \mathcal{H}_X$ are reassembled as

$$S = \sum_{j=0}^3 V_j, \quad D = \sum_{j=0}^3 \|V_j\|^2, \quad G = \|S\|^2, \quad q = G/D,$$

when $D > 0$. The notation follows the exact coherence-to-gain interface in TPC-279 [1]. TPC-280 showed that a raw additive leakage estimate must be normalized using a source lower bound [2].

The present question is narrower: once an arithmetic operator estimate is available, what does it actually buy at the Gate-B output? We assume the typed hypotheses

$$\|A_X\|_{2 \rightarrow 2} \leq KX^{-\sigma}, \quad q \leq Q_X, \quad D \leq d_+ X^a, \tag{1}$$

where $K, d_+ > 0$. The first condition is deliberately stated as an operator statement. It is not inferred from a finite matrix experiment.

2 The exact interface theorem

Theorem 1 (typed arithmetic L^2 interface). *Assume (1). Then*

$$\|A_X S\|_2^2 \leq K^2 X^{-2\sigma} Q_X D. \quad (2)$$

If, in addition,

$$Q_X \leq B X^{-\gamma} + \frac{\ell}{d} X^{-\delta} \leq \left(B + \frac{\ell}{d}\right) X^{-\kappa}, \quad \kappa = \min(\gamma, \delta),$$

then

$$\|A_X S\|_2^2 \leq K^2 d_+ \left(B + \frac{\ell}{d}\right) X^{a-2\sigma-\kappa}. \quad (3)$$

Proof. The operator norm and the definition of G give

$$\|A_X S\|_2^2 \leq \|A_X\|_{2 \rightarrow 2}^2 \|S\|^2 \leq K^2 X^{-2\sigma} G.$$

Since $G = qD \leq Q_X D$, this is (2). The second displayed assumption and $D \leq d_+ X^a$ yield (3) by substitution. $\square$

Corollary 2 (scalar readout). *If $\lambda \in (\ell^2(I_X))^*$ and $\|\lambda\| \leq 1$, then*

$$|\lambda(A_X S)|^2 \leq \|A_X S\|_2^2 \leq K^2 X^{-2\sigma} Q_X D. \quad (4)$$

Proof. Cauchy–Schwarz (or the definition of the dual norm) gives $|\lambda(y)| \leq \|\lambda\| \|y\|_2 \leq \|y\|_2$. $\square$

The theorem is a typed interface rather than a source theorem. In particular, it does not assert that the literal prime/Möbius operator has norm $O(X^{-\sigma})$. It records exactly how such an estimate would enter the existing packet budget. If the inherited margin identity is available, $m^2 = (D/G)m_D^2$, then the TPC-280 compiler contributes $\kappa/2$ to the margin exponent, while the typed operator contributes σ to the output energy. The endpoint ledger therefore still needs the separately paid strict inequality $\sigma - \eta_{\text{eff}} > 1/400$.

3 Attachment is an independent input

The norm estimate in Theorem 1 controls an output from above. It cannot by itself provide a lower bound for a particular scalar attachment. The next proposition makes this non-identifiability exact.

Proposition 3 (equal-norm attachment obstruction). *Let $S = (S_1, S_2) \in \mathbb{R}^2$ be nonzero and put $G = \|S\|^2$. Define Riesz representatives*

$$u_{\parallel} = S, \quad u_{\perp} = (-S_2, S_1),$$

and rank-one functionals $L_u(v) = \langle u, v \rangle$. Then

$$\|L_{u_{\parallel}}\|^2 = \|L_{u_{\perp}}\|^2 = G,$$

whereas

$$L_{u_{\parallel}}(S) = G, \quad L_{u_{\perp}}(S) = 0.$$

Consequently, the packet data (D, G) and the operator norm do not imply any positive lower bound for the squared attachment of S .

Proof. Rotation by 90 degrees preserves the Euclidean norm, so both functional norms squared equal G . Direct calculation gives

$$\langle S, S \rangle = G, \quad \langle (-S_2, S_1), (S_1, S_2) \rangle = 0.$$

The two choices have identical packet geometry and norm data but attachments G^2 and 0 after squaring. This proves the claim. $\square$

Remark 4. *The perpendicular functional is an information-model adversary, not a claim about the actual arithmetic operator. To use Theorem 1 for a prime source one needs a typed identification or nondegeneracy theorem linking the arithmetic readout to the packet direction. Such a theorem is not supplied here.*

4 Exact audit and finite transfer

The release certificate uses four rational packet tuples in $\mathbb{R}^2$. The balanced witness has $(D, G) = (5, 5)$; the near-cancel witness has $(D, G) = (1141/100, 1/100)$; the aligned witness has $(4, 16)$; and the mixed witness has $(14, 2)$. Their normalized ratios are respectively 1, $1/1141$, 4, and $1/7$. For each tuple the parallel and perpendicular functionals have the same squared norm, while the latter has zero attachment.

fixture	D	G	G/D	parallel/perpendicular attachment ²
balanced	5	5	1	25 ; 0
near-cancel	1141/100	1/100	1/1141	1/10000 ; 0
aligned	4	16	4	256 ; 0
mixed	14	2	1/7	4 ; 0

Table 1: Exact packet and attachment audit.

Four finite interface cases instantiate the theorem with $X \in \{8, 16\}$, integer exponents, rational $K, B, \ell/d$, and source upper envelope $D \leq X^2$. The mixed case deliberately has both budget lanes: $B = 3$, $\ell/d = 1$, $\gamma = 1$, $\delta = 2$, so its two-term budget is $49/256$ and its collapsed budget is $1/4$. All comparisons are exact.

As a provenance control, the certificate also copies the twelve coordinate rows from TPC-280. It preserves the eight positive-deficit and four negative-deficit labels and grants zero fixed power. This transfer validates schema and coordinates only; it cannot turn a finite parent table into a growing arithmetic estimate.

5 Claim firewall and conclusion

The main implication is elementary but useful: it tells the next source-level proof exactly what a literal arithmetic L^2 theorem would purchase. The attachment proposition prevents a common category error in which an operator norm or packet cancellation is treated as a lower bound for a selected scalar. The strongest current route statement is therefore

$$\text{packet geometry} \longrightarrow \text{typed arithmetic } L^2 \longrightarrow \text{output energy} \longrightarrow \text{scalar readout},$$

with the middle arrow conditional and the final attachment requiring its own nondegeneracy input.

The literal source arithmetic L^2 estimate, typed attachment theorem, full Gate B, and the twin-prime conclusion remain open. The fixed-power credit is zero. The Session-named Route-A and Route-B evaluator files are not present in the checkout; the local proof package, exact certificate, independent replay, hostile stress audit, and fail-closed Bridge-B checker provide the scoped release evaluation.

References

- [1] Liang Wang, “A Minimal Coherence-to-Gain Criterion for Four-Packet Reassembly,” TPC-279 project release, 2026.
- [2] Liang Wang, “An Additive-Leakage Compiler for Signed Gain and Endpoint Budgets,” TPC-280 project release, 2026.